1 Sparse-View Wheat Field Reconstruction with Gaussian Splatting and Generative Repair

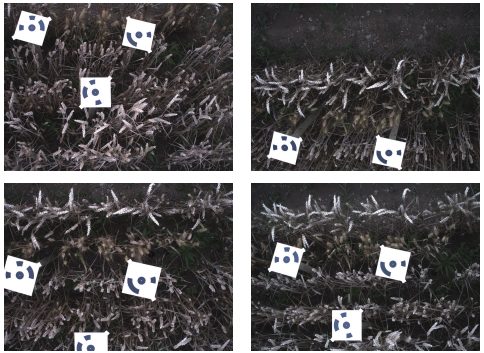
Gavin Gleinig

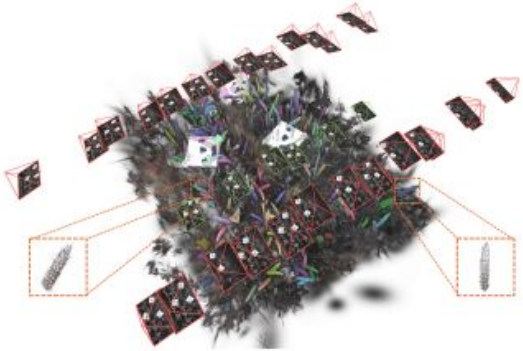
Motivation: 3D Reconstruction in Agriculture

Goal: Create accurate 3D wheat models for agricultural phenotyping (HTFP), which is crucial for plant breeding

Challenge: Complex geometry, occlusion, and *sparse data* hinder traditional methods and standard 3D Gaussian Splatting (3DGS).

My Idea: Leverage powerful 2D generative priors (diffusion models) to fill gaps and correct artifacts left by sparse data.



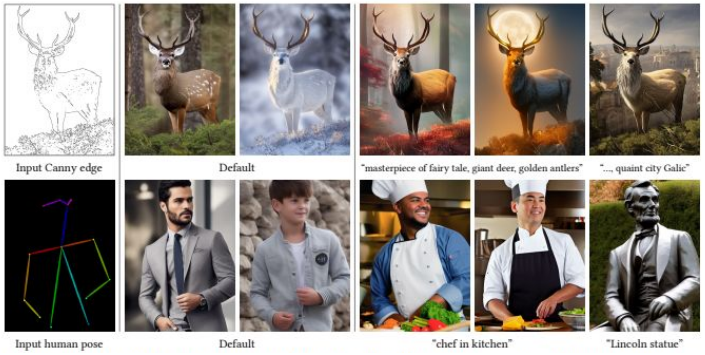


AI Tools: 3DGS & Generative Models

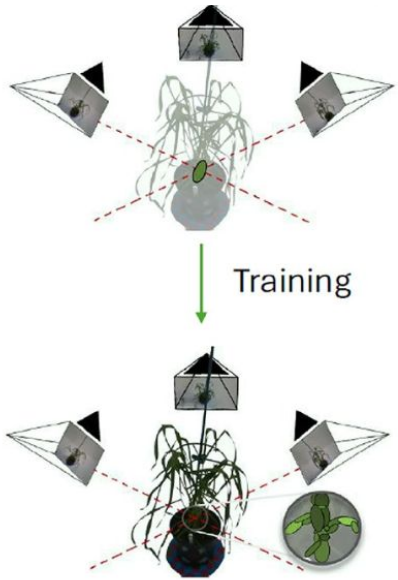
3D Gaussian Splatting (3DGS): A differentiable rasterization technique: Represents a 3D scene as a collection of 3D Gaussian shapes, allowing for fast, photorealistic synthesis of novel views.

Generative Diffusion Models: Learn rich visual priors from vast datasets; excel at realistic image generation/enhancement.

Combine 3DGS's spatial representation with 2D models' priors to overcome data limitations.



Adding Conditional Control to Text-to-Image Diffusion Models



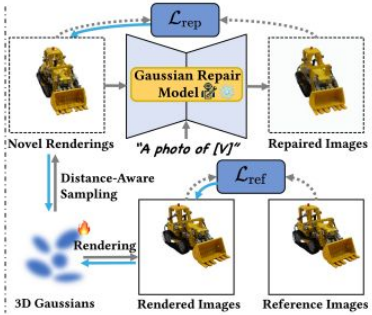
High-fidelity wheat plant reconstruction using 3D Gaussian splatting and neural radiance fields

Related Work: Enhancing 3DGS with 2D Priors

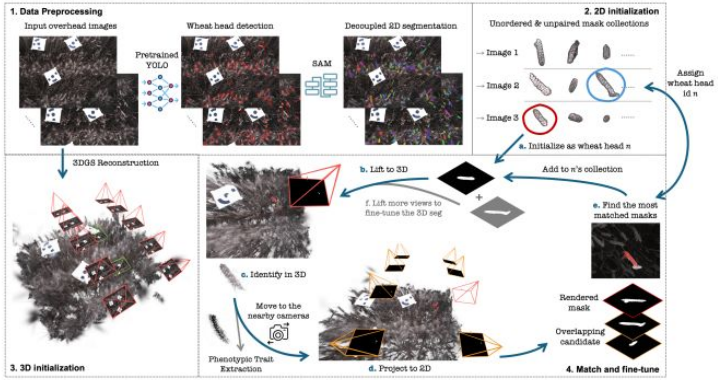
Active research area integrates 2D priors into 3DGS. Key approaches are:

- *Generative Augmentation and Guidance:* Use diffusion outputs in the 3DGS loss (e.g. LM-Gaussian, 3DGS-Enhancer) or use diffusion-enhanced views as new training data (e.g., AugGS)
- *3D Segmentation*: Utilize 2D segmentation priors, like masks or features from models like SAM, to segment 3D scenes represented by Gaussian Splatting. (e.g., SAGA, GaussianCut)

Current agriculture applications: Wheat3DGS, PlantDreamer
No applications of 3DGS refinement with generative diffusion models in the agriculture field

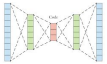


GaussianObject: High-Quality 3D Object Reconstruction from Four Views with Gaussian Splatting



Wheat3DGS: In-field 3D Reconstruction, Instance Segmentation and Phenotyping of Wheat Heads with Gaussian Splatting

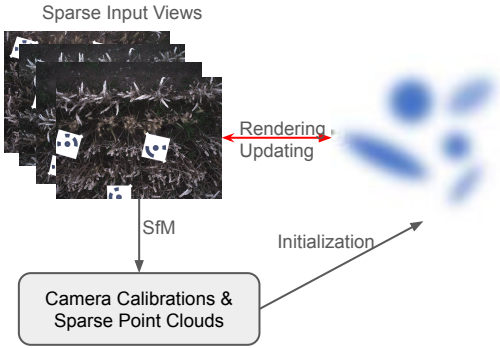
Method: Overview

My Approach: Refinement of sparse-view 3DGS using a scene-specific "Repair Model". 

Core Idea: Use enhanced renderings as additional training views during 3D reconstruction .

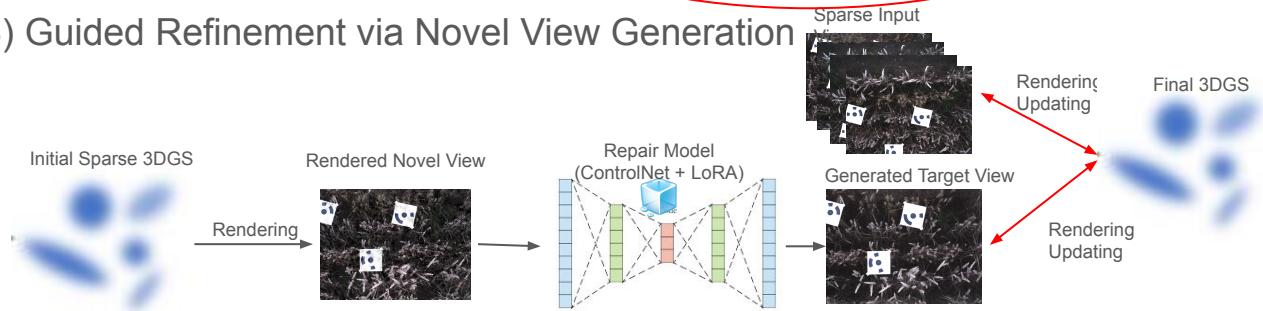
Repair Model: ControlNet + Diffusion Model + Fine-tuned LoRA layers.

1) Initial Sparse Reconstruction



2) Fine-Tuning the Scene-Specific Repair Model

3) Guided Refinement via Novel View Generation



Method: LoRA ControlNet “Repair Model”

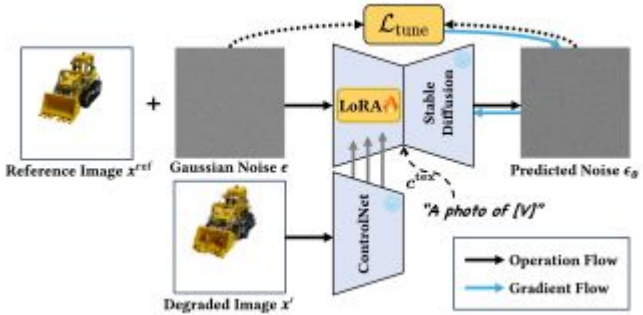
Function: Take blurry, lower-detail, or artifact-heavy images rendered from the current state of the 3D Gaussian Splatting (3DGS) model (especially from virtual viewpoints) and enhance them into sharp, realistic images

Architecture: ControlNet conditions generation on input rendering structure. LoRA layers adapt a pre-trained diffusion model.

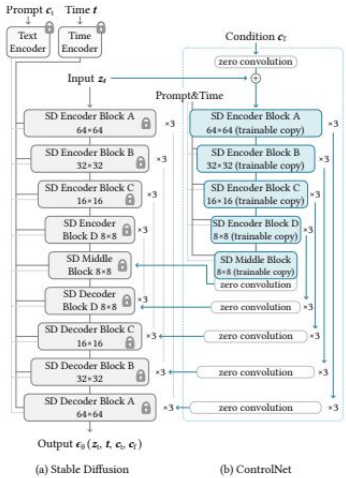
Add LoRA layers to all embedding, linear, and convolution layers of:

- Diffusion U-Net
- ControlNet U-Net,
- CLIP model.

Scene Adaptation: LoRA layers fine-tuned on the *original sparse input images*



GaussianObject: High-Quality 3D Object Reconstruction from Four Views with Gaussian Splatting



Adding Conditional Control to Text-to-Image Diffusion Models

Data, Baseline 3DGS Results

Dataset: Wheat3DGS

- RGB images for seven wheat plots,
- calibrated camera poses,
- laser scans,
- view-consistent segmentation masks

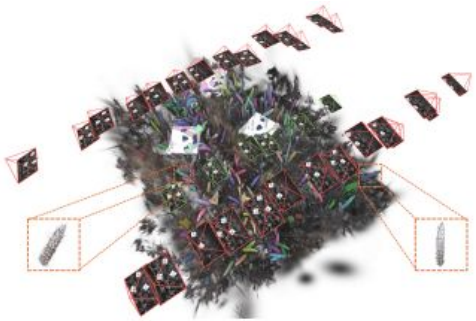


Figure 1. 3D Gaussian Splatting reconstruction of a wheat plot with segmented 3D wheat heads instances (in different colors). We use 30 views for reconstruction (red frustums) and 6 out-of-distribution views for evaluation (green frustums).

Blurriness, geometric errors, and missing details are present when training with sparse views

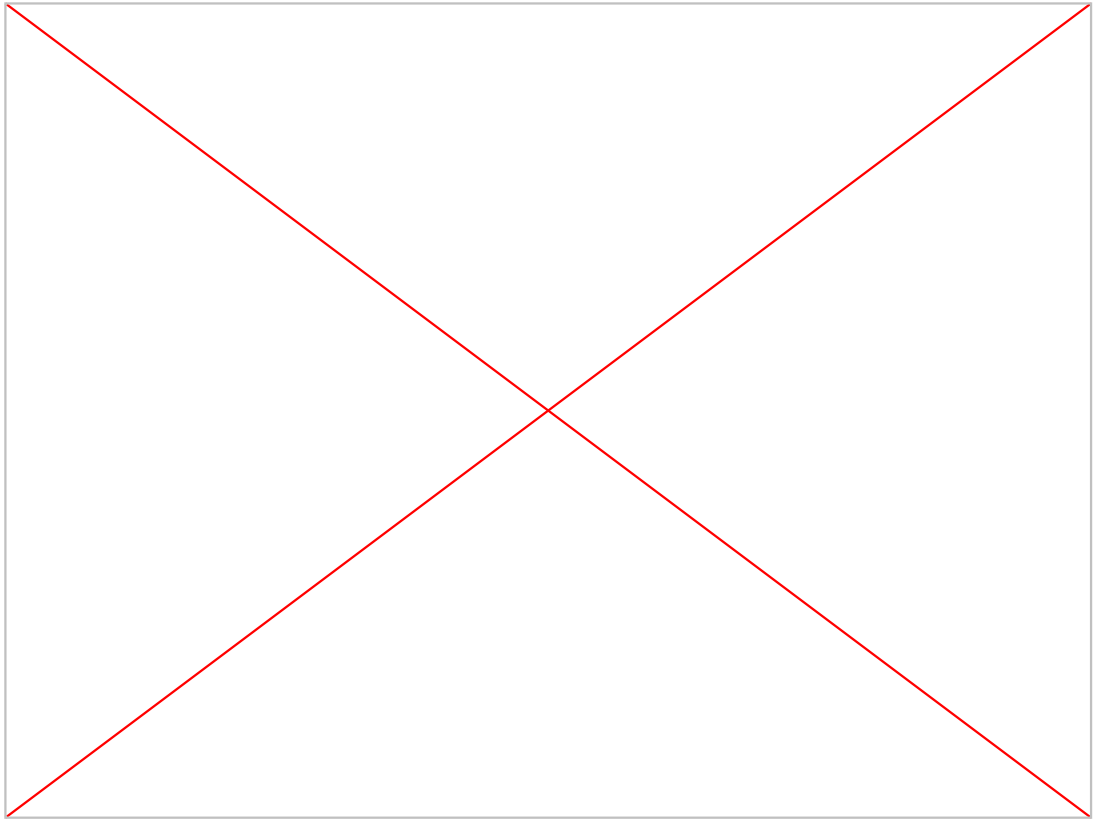
Novel / Test Set View

Ground Truth

Sparse (5 Views)

Dense (30 Views)

Sparse Results Continued



Expected Result and Current Work

Expected: Significantly improved visual fidelity and geometric completeness compared to baseline sparse 3DGS.

Current Work:

- Complete LoRA of Repair Model
- Evaluate and compare with Dense Views

```
  | Name              | Type               | Params | Mode
-----------------------------------------------------------------
0 | model             | DiffusionWrapper   | 865 M  | train
1 | first_stage_model | AutoencoderKL      | 83.7 M | eval
2 | cond_stage_model  | FrozenCLIPEmbedder | 126 M  | eval
3 | control_model     | ControlNet         | 364 M  | train
-----------------------------------------------------------------
1.2 B     Trainable params
206 M     Non-trainable params
1.4 B     Total params
5,758.887 Total estimated model params size (MB)
2104      Modules in train mode
439       Modules in eval mode
```

Conclusion

My project addresses sparse-view 3DGS for complex wheat scenes via dataset augmentation using a tailored generative Repair Model.

Questions?